

```

const Inventory = (() => {
const grid = document.querySelector("#inventory-grid");
const inventoryPanel = document.querySelector("#inventory");
const backpackButton = document.querySelector("#backpack-button");

const state = {
items: [],
selectedItem: null
};

const render = () => {
grid.innerHTML = "";
const slots = 6;
for (let i = 0; i < slots; i += 1) {
const slot = document.createElement("button");
slot.className = "inventory-slot";
slot.type = "button";
const item = state.items[i];
slot.textContent = item ? item.name : "Empty";
if (item && state.selectedItem?.id === item.id) {
slot.classList.add("selected");
}
slot.addEventListener("click", () => {
state.selectedItem = item || null;
render();
});
grid.appendChild(slot);
}
};

const addItem = (item) => {
if (state.items.length >= 6) {
return;
}
state.items.push(item);
render();
};

const removeItem = (itemId) => {
state.items = state.items.filter((item) => item.id !== itemId);
}

```

```
if (state.selectedItem?.id === itemId) {
  state.selectedItem = null;
}
render();
};

const toggle = () => {
  inventoryPanel.classList.toggle("open");
};

backpackButton.addEventListener("click", toggle);

return {
  addItem,
  removeItem,
  render,
  get selectedItem() {
    return state.selectedItem;
  }
};
})();
```